

Lab - Running the Container on an Azure VM

To Begin with the Lab

Summary

This lab demonstrates how to deploy a Docker container from Azure Container Registry (ACR) onto an Azure Linux Virtual Machine. The process includes creating an Ubuntu VM, installing the Docker Engine, logging into ACR, pulling the Docker image, and running it as a container.

Understanding

An Azure Virtual Machine can act as a Docker host by installing the Docker Engine. Instead of storing images locally, the VM pulls Docker images from Azure Container Registry (ACR). Once the image is downloaded, it can be started as a container with port mapping, allowing users to access the application through the VM's public IP address.

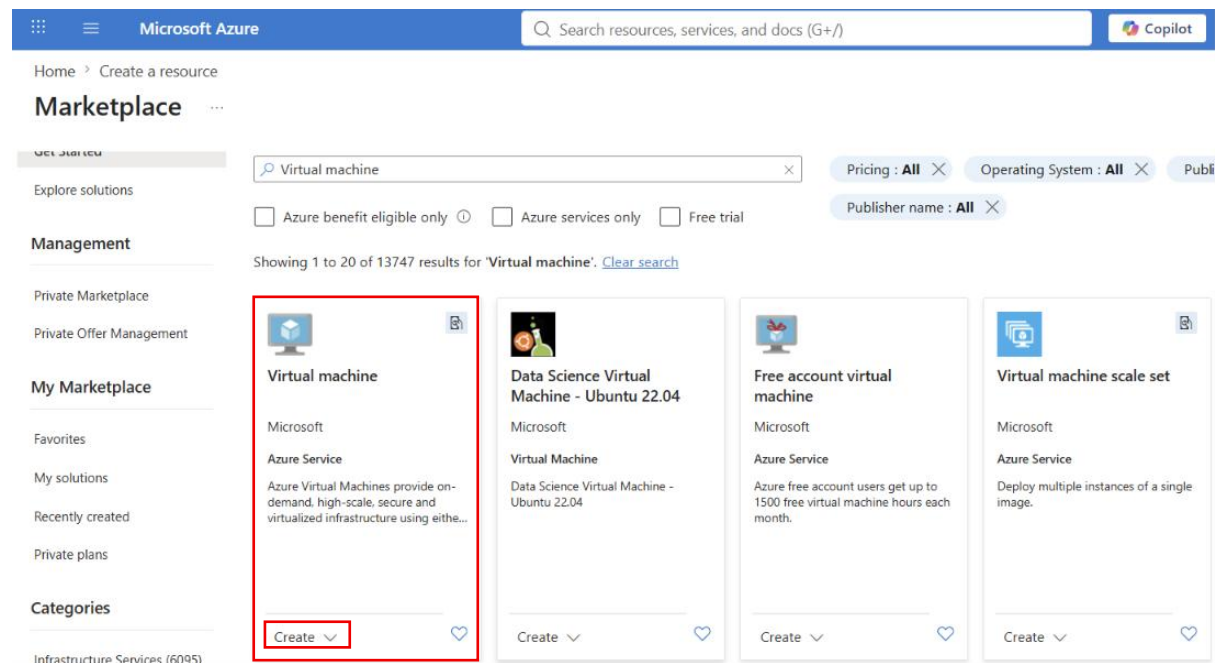
Example:

- Docker Image: regdeves12.azurecr.io/dockerapp:v1
- VM Public Port: 80
- Container Port: 80
- Access URL: <http://<VM-Public-IP>>

Lab Steps

Step 1: Create an Azure Virtual Machine

- Open the **Azure Portal**.
- Navigate to **Virtual Machines**.
- Click **Create** → **Azure Virtual Machine**.



- Select your **Resource Group**.
- Enter a unique **Virtual Machine Name**.
- Choose **Region** (e.g., East Asia).

- Choose **Availability options** – **NO infrastructure redundancy required**.
- Select **Ubuntu Server 24.04 LTS**.

Microsoft Azure

Home > Create a resource > Marketplace

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Virtual machine name *

Region * [Deploy to an Azure Extended Zone](#)

Availability options

Security type [Configure security features](#)

Image *

< Previous | Next : Disks > | Review + create

- Keep the default size (**Standard_D2ls**).
- Choose **Password Authentication**.
- Enter the **username** and **password**.

Microsoft Azure

Home > Create a resource > Marketplace

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

VM architecture ☐ Arm64 ☒ x64

Run with Azure Spot discount ☐

Size * [See all sizes](#) | [Customize cores](#)

Enable Hibernation ☐ Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#)

Administrator account

Authentication type ☐ SSH public key ☒ Password

Username *

Password *

Confirm password *

< Previous | Next : Networking > | Review + create

- Allow **HTTP (Port 80)** inbound traffic.
- Leave the remaining settings as default.

- Click **Review + Create**.
- Click **Create**.

Username * ⓘ ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ ☐ None ☒ Allow selected ports

Select inbound ports *

✓

☒ HTTP (80)
 ☐ HTTPS (443)
 ☒ SSH (22)

< Previous Next : Disks > **Review + create**

Step 2: Connect to the Virtual Machine

- Open the created VM.
- From the left menu, select **Serial Console**.

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home

docker-vm Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Resource health

Boot diagnostics

Serial console

Reset password

docker-vm virtual machine agent status is not ready. Troubleshoot the issue →

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback CL

Essentials

Resource group (move) : [Webapp](#)

Status : Running

Location : East Asia

Subscription (move) : [Visual Studio Enterprise Subscription](#)

Subscription ID : 294539c9-c109-4db8-b785-0043716c0b11

Operating system : Linux

Size : Standard D2ls v5 (2 vcpus, 4 GiB memory)

Primary NIC public IP : [23.101.5.38](#) [1 associated public IPs](#)

Virtual network/subnet : [docker-vm-vnet/default](#)

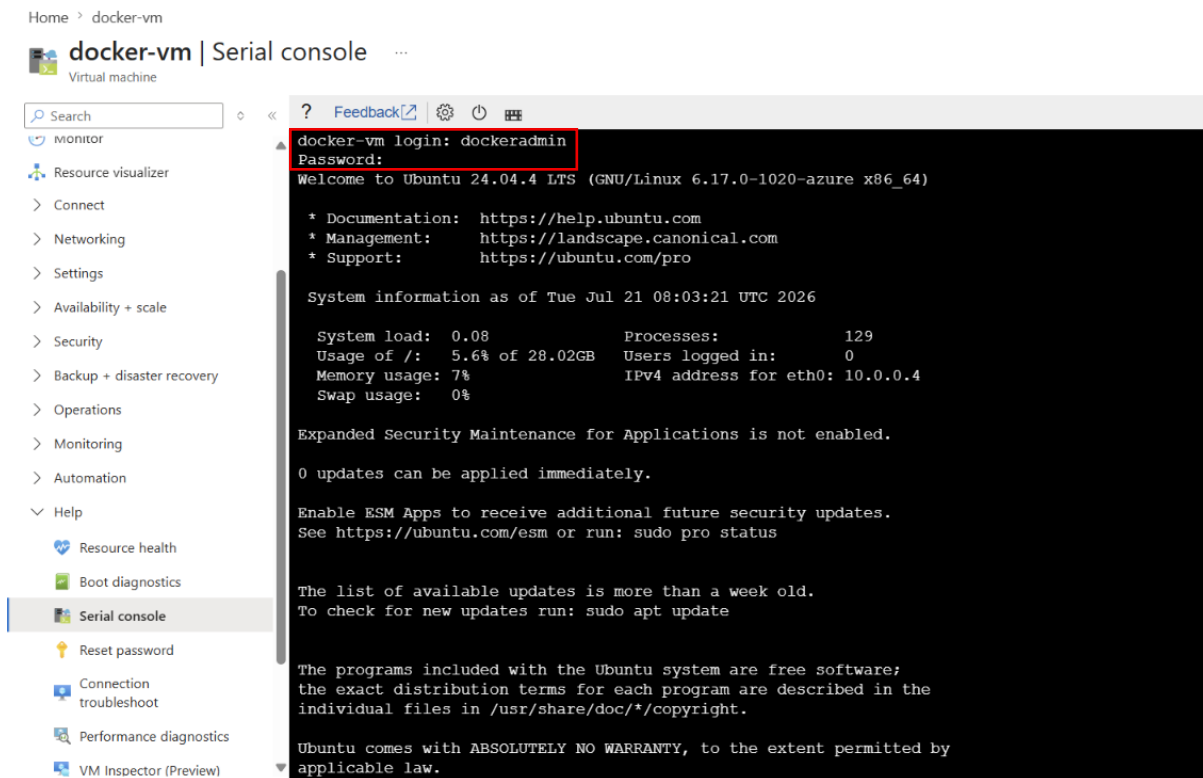
DNS name : [Not configured](#)

Health state : -

Time created : 7/21/2026, 7:23 AM UTC

Taas (edit) Add taas

- Press **Enter** if required.
- Log in using the VM **username** and **password**.

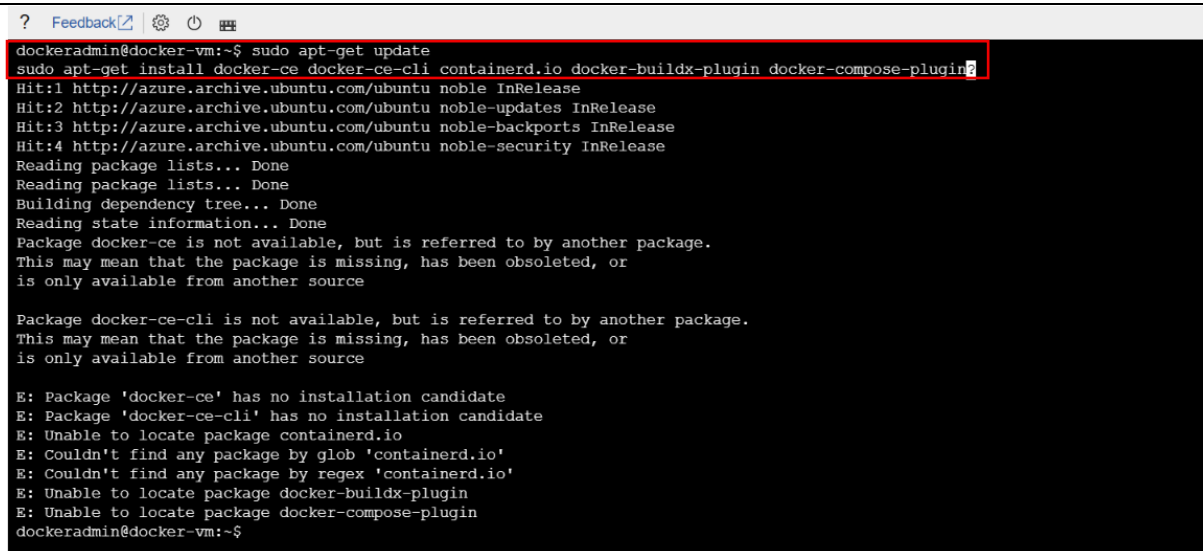


Step 3: Install Docker Engine

- Open the official Docker installation instructions for Ubuntu.
- Copy the commands to:
 - Add Docker's GPG key.
 - Add the Docker repository.
 - Update package information.
- Run the installation command to install Docker Engine.

```

sudo apt-get update
sudo apt install -y docker.io
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-
compose-plugin
  
```



Step 4: Verify Docker Installation

Check that Docker is installed successfully.

`docker --version`

or

`sudo docker --version`

```
? Feedback [?] [?] [?] [?]
dockeradmin@docker-vm:~$ sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package docker-ce is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

Package docker-ce-cli is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'docker-ce' has no installation candidate
E: Package 'docker-ce-cli' has no installation candidate
E: Unable to locate package containerd.io
E: Couldn't find any package by glob 'containerd.io'
E: Couldn't find any package by regex 'containerd.io'
E: Unable to locate package docker-buildx-plugin
E: Unable to locate package docker-compose-plugin
dockeradmin@docker-vm:~$ docker --version
Command 'docker' not found, but can be installed with:
sudo apt install docker.io      # version 29.1.3-0ubuntu3~24.04.2, or
sudo apt install podman-docker  # version 4.9.3+ds1-1ubuntu0.2
dockeradmin@docker-vm:~$
```

Step 5: Login to Azure Container Registry

Use Docker to authenticate with **your Azure Container Registry**.

```
sudo docker login <>.azurecr.io
```

Example:

```
sudo docker login regdeves12.azurecr.io
```

- Enter the **Username**.
- Enter the **Password**.
- Verify the message:

Login Succeeded

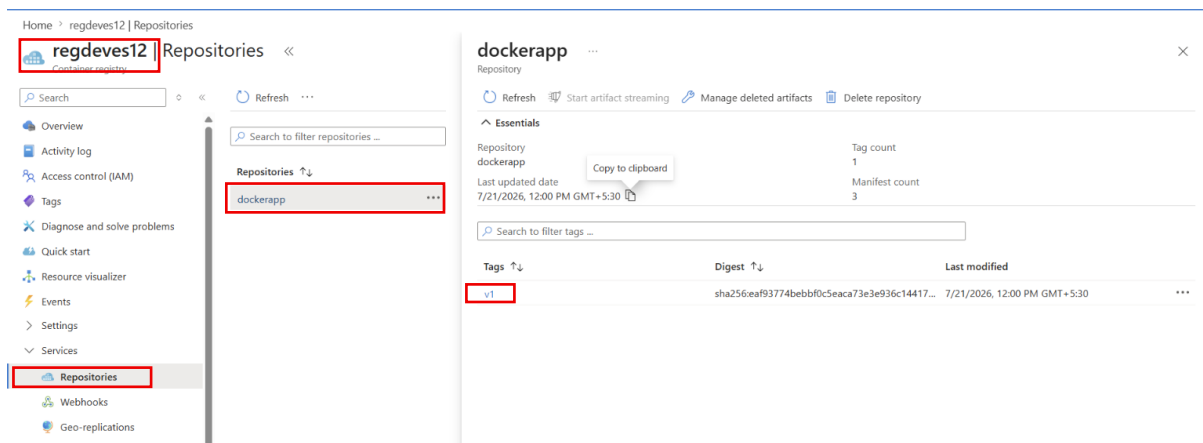
```
? Feedback [?] [?] [?] [?]
dockeradmin@docker-vm:~$ sudo docker login regdeves12.azurecr.io
Username: regdeves12
Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

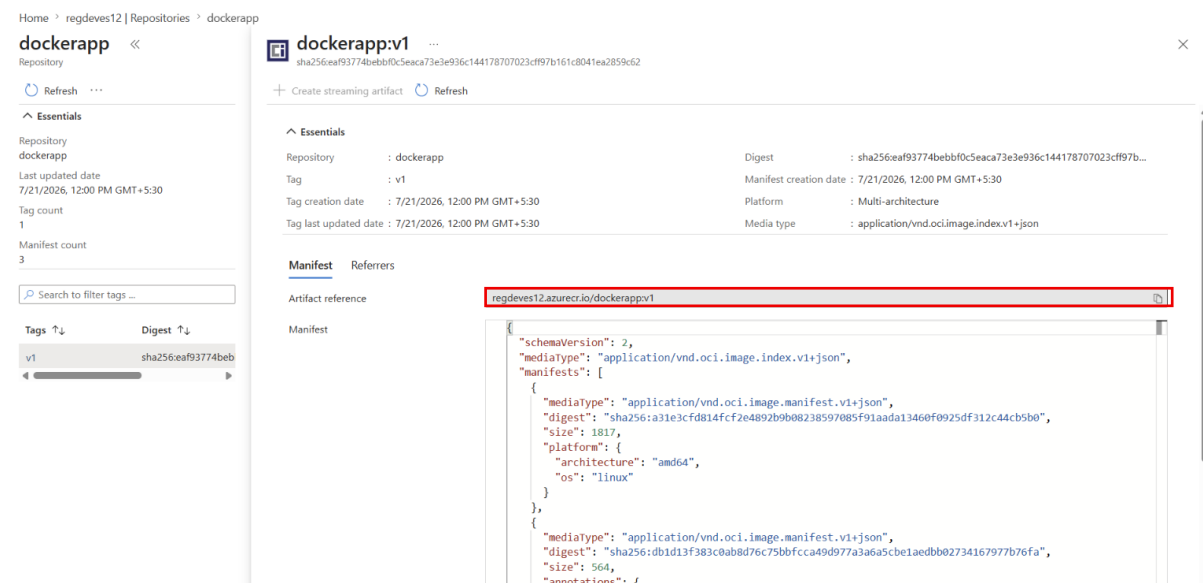
Login Succeeded
dockeradmin@docker-vm:~$
```

Step 6: Get the Image Name

- Open **Azure Container Registry**.
- Select **Repositories**.
- Open the **dockerapp** repository.
- Select the **v1** tag.



- Copy the full image URL.



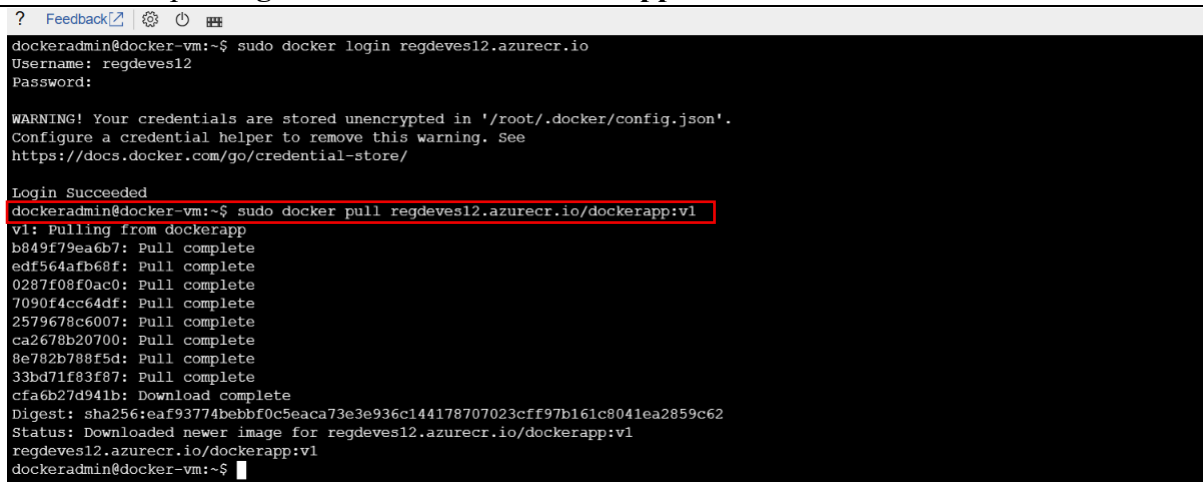
Example:

regdevs12.azurecr.io/dockerapp:v1

Step 7: Pull the Docker Image

Download the image from Azure Container Registry.

```
sudo docker pull regdevs12.azurecr.io/dockerapp:v1
```



Step 8: Verify the Downloaded Image

List the Docker images available on the VM.

```
sudo docker images
```

Confirm that the dockerapp:v1 image is present.

```
? Feedback ? ? ?
dockeradmin@docker-vm:~$ sudo docker images
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
regdeves12.azurecr.io/dockerapp:v1 eaf93774bebb 359MB 101MB
dockeradmin@docker-vm:~$
```

Step 9: Run the Docker Container

Start a container using port mapping.

```
sudo docker run -d -p 80:80 regdeves12.azurecr.io/dockerapp:v1
```

Explanation:

- d → Run the container in the background.
- p 80:80 → Map VM Port 80 to Container Port 80.

```
? Feedback ? ? ?
dockeradmin@docker-vm:~$ sudo docker run -d -p 80:80 regdeves12.azurecr.io/dockerapp:v1
64233f63f63bdf0b26c7fed154797d232b83b072a1c6fb0790bb4a71943925a
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint goofy_pike (e0ce5998559729bef3cf0efa4114e4bfc2abe6c4c39cb196aaf71e5c0ce3cf): Bind for 0.0.0.0:80 failed: port is already allocated

Run 'docker run --help' for more information
dockeradmin@docker-vm:~$
```

Step 10: Verify the Running Container

Check that the container is running.

```
sudo docker ps
```

```
? Feedback ? ? ?
dockeradmin@docker-vm:~$ sudo docker run -d -p 80:80 regdeves12.azurecr.io/dockerapp:v1
64233f63f63bdf0b26c7fed154797d232b83b072a1c6fb0790bb4a71943925a
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint goofy_pike (e0ce5998559729bef3cf0efa4114e4bfc2abe6c4c39cb196aaf71e5c0ce3cf): Bind for 0.0.0.0:80 failed: port is already allocated

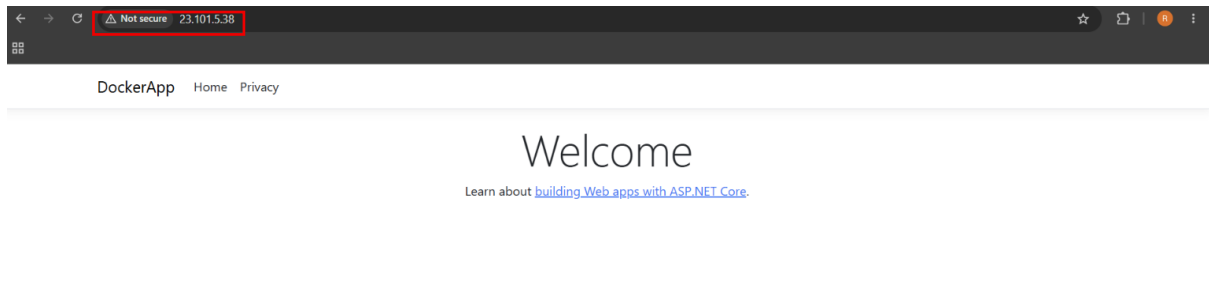
Run 'docker run --help' for more information
dockeradmin@docker-vm:~$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
bf929ad955cd regdeves12.azurecr.io/dockerapp:v1 "dotnet DockerApp.dll" 3 minutes ago Up 3 minutes 0.0.0.0:80->80/tcp, [::]:80->80/tcp, 8080/tcp
funny_cohen
dockeradmin@docker-vm:~$
```

Step 11: Access the Application

- Copy the VM's Public IP Address.

The screenshot shows the Azure portal interface for a virtual machine named 'docker-vm'. The 'Overview' tab is selected, displaying various details about the VM. A red box highlights the 'Primary NIC public IP' field, which shows the IP address '23.101.5.38'. Other visible details include the operating system 'Linux (ubuntu 24.04)', the size 'Standard_D2ls_v5', and the subscription 'Visual Studio Enterprise Subscription'.

- Paste a web browser.



The application running inside the Docker container should be displayed.